Linear Algebra and Its Applications

Section 6.6: Applications to Linear Models

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.6 Applications to linear models

A model is linear in its *unknown parameters*, even if it contains x^2, $\sin x$, or $e^{-0.02x}$. For observations (x_i, y_i), write the predicted value as a row of a design matrix X times a parameter vector β. The observation equation is $y = X\beta + \varepsilon$. Least squares minimizes $\|\varepsilon\|$, and the fitted parameters solve $X^T X\widehat{\beta} = X^T y$. As a check, the fitted residual always satisfies $X^T(y - X\widehat{\beta}) = 0$. We retain unrounded coefficients in computations and round only displayed predictions.

Practice problem 6.6.P1: A seasonal term is still linear in its coefficient

For the proposed seasonal model, use

$$X = \begin{bmatrix} 1 & x_1 & \sin(2\pi x_1/12) \\ \vdots & \vdots & \vdots \\ 1 & x_n & \sin(2\pi x_n/12) \end{bmatrix}, \quad \beta = (\beta_0, \beta_1, \beta_2)^T, \quad y = (y_1, \ldots, y_n)^T.$$

The ith product row equals the predicted sales at month x_i. The sine values are known once the observation times are supplied; only their coefficients are unknown, so this is a linear least-squares model.

Exercise 6.6.1: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{9}{10} \\ \frac{2}{5} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{9}{10} + (\frac{2}{5})x}.$$

The residual $\begin{bmatrix} \frac{1}{10} \\ -\frac{3}{10} \\ \frac{3}{10} \\ -\frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{5}$.

Exercise 6.6.2: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 12 \\ 12 & 46 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 25 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} -\frac{3}{5} \\ \frac{7}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = -\frac{3}{5} + (\frac{7}{10})x}.$$

The residual $\begin{bmatrix} -\frac{1}{10} \\ \frac{1}{5} \\ -\frac{1}{5} \\ \frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{10}$.

Exercise 6.6.3: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 4 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{11}{10} \\ \frac{13}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{11}{10} + (\frac{13}{10})x}.$$

The residual $\begin{bmatrix} \frac{1}{5} \\ -\frac{1}{10} \\ -\frac{2}{5} \\ \frac{3}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{3}{10}$.

Exercise 6.6.4: Fit a line through the normal equations

Rows of X are $(1, x_i)$, in the order of the stated data. Thus

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}, \qquad y = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 0 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 4 & 16 \\ 16 & 74 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}.$$

Reducing this two-variable system gives $\widehat{\beta} = \begin{bmatrix} \frac{43}{10} \\ -\frac{7}{10} \end{bmatrix}$, so the fitted line is

$$\boxed{y = \frac{43}{10} + (-\frac{7}{10})x}.$$

The residual $\begin{bmatrix} \frac{1}{10} \\ -\frac{1}{5} \\ \frac{1}{5} \\ -\frac{1}{10} \end{bmatrix}$ has zero sum and zero dot product with the observed x-coordinate list. These are exactly the two normal equations, verifying the minimizer. Its squared error is $\frac{1}{10}$.

Exercise 6.6.5: Exactly when does a line fit have unique parameters?

By Section 6.5, the normal equations have a unique solution exactly when the two columns of X are independent. They are the all-ones column and $(x_1, \ldots, x_n)^T$. If all x_i equal the same c, the second column is c times the first, so independence and uniqueness fail. Conversely, suppose $ax_i + b = 0$ at all data points and two abscissas differ. Subtracting the equations at those two points forces $a = 0$, then $b = 0$. Thus the columns are independent precisely when at least two distinct x-coordinates occur. Repeated observations at one x cannot identify both an intercept and a slope uniquely.

Exercise 6.6.6: Three distinct abscissas identify a quadratic fit

The three design columns are $1, x, x^2$. A dependence among them would give a polynomial $c_0+c_1t+c_2t^2$ that vanishes at every observed abscissa, in particular at three distinct ones. A nonzero polynomial of degree at most two cannot have three distinct roots. Hence all coefficients must be zero, proving independence of the columns. The full-column-rank least-squares theorem then gives a unique best-fitting parameter vector. The fitted polynomial may happen to have zero quadratic coefficient; uniqueness concerns the polynomial, not a requirement that its degree be exactly two.

Exercise 6.6.7: Fit a quadratic with no constant term

(a) Put the known powers into the design columns.

For the five observations, the model is

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix}, \quad y = (1.8, 2.7, 3.4, 3.8, 3.9)^T, \quad \beta = (\beta_1, \beta_2)^T.$$

There is no column of ones, because the proposed model has no intercept.

(b) Solve and state the curve.

The exact normal equations are

$$\begin{bmatrix} 55 & 225 \\ 225 & 979 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 521/10 \\ 403/2 \end{bmatrix}.$$

Their solution is $\widehat{\beta} = (14171/8050, -32/161)^T$. Thus the least-squares curve is

$$y = \frac{14171}{8050}x - \frac{32}{161}x^2 \approx 1.760373x - 0.198758x^2.$$

Substitution in the normal equations verifies both residual moments are zero. This certifies the fit with the supplied decimal data treated exactly.

Exercise 6.6.8: Fit a cubic cost model without fixed costs

(a) Record the general model.

Use $X_{i,:} = (x_i, x_i^2, x_i^3)$, $y = (y_1, \ldots, y_n)^T$, and $\beta = (\beta_1, \beta_2, \beta_3)^T$. Again no column of ones appears. The residual equation is $y = X\beta + \varepsilon$.

(b) Use the eight supplied observations.

Here $x_i = 4, 6, 8, 10, 12, 14, 16, 18$, and the observed costs are $1.58, 2.08, 2.50, 2.80, 3.10, 3.40, 3.80, 4.32$, in the stated thousand-unit scales. The normal equations are

$$\begin{bmatrix} 1136 & 16192 & 245312 \\ 16192 & 245312 & 3866368 \\ 245312 & 3866368 & 62617856 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 7254/25 \\ 100636/25 \\ 1501896/25 \end{bmatrix}.$$

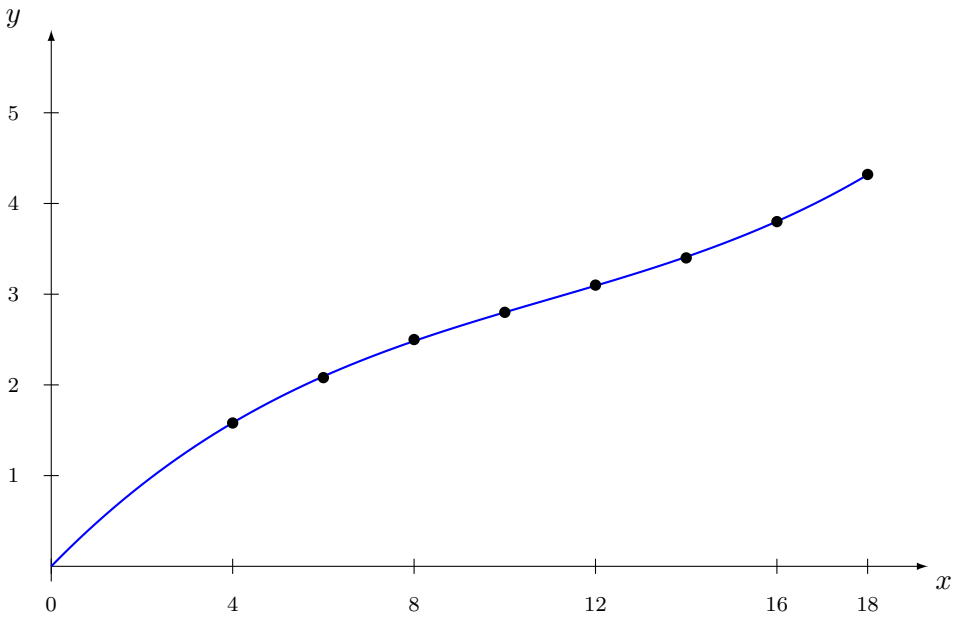
Solving with exact rational data gives

$$\widehat{\beta} = (8970503/17479000,\ -319181/9534000,\ 15221/14982000)^T.$$

The fitted curve is therefore

$$y \approx 0.513216031x - 0.0334781833x^2 + 0.00101595248x^3.$$

The exact residual satisfies $X^T(y - X\widehat{\beta}) = 0$. The graph below shows the fitted curve in blue and all eight observations as black points; both axes use the problem's thousand-unit scale.



Exercise 6.6.9: A trigonometric model linear in two coefficients

Use

$$X = \begin{bmatrix} \cos 1 & \sin 1 \\ \cos 2 & \sin 2 \\ \cos 3 & \sin 3 \end{bmatrix}, \quad y = (7.9, 5.4, -0.9)^T, \quad \beta = (A, B)^T.$$

Angles are in radians. Although the basis functions are trigonometric, the unknowns A, B enter linearly. Minimize $\|y - X\beta\|$ by solving $X^T X\widehat{\beta} = X^T y$, or use QR on X directly. A numerical solution gives $A \approx 2.34212062$, $B \approx 7.44750155$, so $y \approx 2.34212062\cos x + 7.44750155\sin x$. Using unrounded coefficients, the normal residual is below 3×10^{-14} in absolute value, a check on the computed fit.

Exercise 6.6.10: Estimate two initial radioactive masses

(a) Treat the decay functions as known columns.

At times $t_i = 10, 11, 12, 14, 15$, use

$$X_{i,:} = (e^{-0.02t_i}, e^{-0.07t_i}), \quad \beta = (M_A, M_B)^T,$$

with observation vector $(21.34, 20.68, 20.05, 18.87, 18.30)^T$. The decay constants are supplied, so the two masses, not the exponent rates, are the unknown linear parameters.

(b) Fit the two masses.

Numerical least squares gives $M_A \approx 19.94109495$ and $M_B \approx 10.10151413$ grams. Thus the fitted curve is

$$y(t) \approx 19.94109495e^{-0.02t} + 10.10151413e^{-0.07t}.$$

Both fitted masses are positive. The residual norm is approximately 0.0114745 grams, and the normal residual computed from unrounded coefficients is below 3×10^{-14}. The residual is not assumed to vanish, because the measured data are rounded.

Exercise 6.6.11: Linearize the comet equation in the correct parameters

The relation $r = \beta + e(r\cos\theta)$ is linear in β, e. For the given observations, use design rows $(1, r_i\cos\theta_i)$, not $(1, \cos\theta_i)$, and observation vector (r_i). Least squares gives $\beta \approx 1.45092922$ and $e \approx 0.81105251$; the fitted normal residual is below 2×10^{-14}. Since $0 < e < 1$, the predicted orbit is an ellipse. For a new angle, solve the original model for its new radius:

$$r = \frac{\beta}{1 - e\cos\theta}, \qquad r(4.6) \approx 1.32995445.$$

The denominator is positive here (indeed $e < 1$ makes it positive at every angle).

Conclusion. An elliptic orbit, with predicted radius about 1.330 in the same units as the observed radii.

Exercise 6.6.12: Regress pressure on logarithmic weight

Use design rows $(1, \ln w_i)$ and observation vector $(91, 98, 103, 110, 112)^T$. If the printed rounded logarithms $3.78, 4.11, 4.39, 4.73, 4.88$ are used, the fitted line is

$$p \approx 18.56421016 + 19.24070120\ln w.$$

At $w = 100$, using $\ln 100 \approx 4.605170186$, this predicts $p \approx 107.17091367$ mmHg. For clarity about precision: recomputing the logarithms directly from $w = 44, 61, 81, 113, 131$ instead of using their rounded table entries gives $p \approx 17.92434012 + 19.38499925\ln w$, and a prediction 107.19556074 mmHg. Both computations satisfy their respective normal equations to roundoff.

Conclusion. About 107 mmHg. The small difference between the two more detailed predictions comes from rounding the predictor data.

Exercise 6.6.13: Fit takeoff position and differentiate the fitted polynomial

(a) Build the cubic design matrix from the thirteen times.

For $t = 0, 1, \ldots, 12$, take row $(1, t, t^2, t^3)$ and the given position vector in feet. The normal equations have coefficient matrix

$$X^TX = \begin{bmatrix} 13 & 78 & 650 & 6084 \\ 78 & 650 & 6084 & 60710 \\ 650 & 6084 & 60710 & 630708 \\ 6084 & 60710 & 630708 & 6735950 \end{bmatrix},$$

with right side $X^Ty = (19001/5, 351277/10, 3480639/10, 35998009/10)^T$. Solving gives the exact coefficient vector

$$\widehat{\beta} = (-89/104,\ 75315/16016,\ 222437/40040,\ -313/11440)^T.$$

Thus, approximately,

$$y(t) = -0.855769231 + 4.702485015t + 5.555369630t^2 - 0.02736013986t^3.$$

The exact coefficients make $X^T(y - X\widehat{\beta}) = 0$; distinct times ensure the four design columns are independent, so the fit is unique.

(b) Differentiate before evaluating.

The fitted velocity is $y'(t) = \widehat{\beta}_1 + 2\widehat{\beta}_2 t + 3\widehat{\beta}_3 t^2$. At $t = 4.5$ this gives $\boxed{53.03868319 \text{ feet/second}}$. This estimates instantaneous velocity from the fitted curve, rather than dividing the fitted position by the elapsed time.

Exercise 6.6.14: The fitted line passes through the mean point

The residual $r = y - X\widehat{\beta}$ is orthogonal to each column of X, including the all-ones column. Thus its entries sum to zero: $\sum_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0$. Divide by $n > 0$ and use the definitions of the means to obtain

$$\overline{y} = \widehat{\beta}_0 + \widehat{\beta}_1 \overline{x}.$$

This says exactly that $(\overline{x}, \overline{y})$ lies on the fitted line.

Exercise 6.6.15: Derive the scalar normal equations

For design columns $\mathbf{1}$ and x, the four Gram entries are $\mathbf{1}^T\mathbf{1} = n$, $\mathbf{1}^T x = x^T\mathbf{1} = \sum x_i$, and $x^T x = \sum x_i^2$. Similarly, $X^T y = (\sum y_i, \sum x_i y_i)^T$. Multiplication of the resulting two-by-two system gives

$$n\widehat{\beta}_0 + \widehat{\beta}_1 \sum x_i = \sum y_i, \qquad \widehat{\beta}_0 \sum x_i + \widehat{\beta}_1 \sum x_i^2 = \sum x_i y_i,$$

which are precisely equations (7).

Exercise 6.6.16: Invert the two-by-two normal matrix

Write $S_x = \sum x_i$, $S_{xx} = \sum x_i^2$, $S_y = \sum y_i$, and $S_{xy} = \sum x_i y_i$. When at least two abscissas differ, $D = nS_{xx} - S_x^2 = n\sum_i (x_i - \overline{x})^2 > 0$. Thus the normal matrix inverse is

$$\frac{1}{D}\begin{bmatrix} S_{xx} & -S_x \\ -S_x & n \end{bmatrix}.$$

Multiplying it by $(S_y, S_{xy})^T$ gives

$$\boxed{\widehat{\beta}_0 = \frac{S_{xx}S_y - S_x S_{xy}}{D}, \qquad \widehat{\beta}_1 = \frac{nS_{xy} - S_x S_y}{D}}.$$

Equivalently $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$. If all abscissas coincide, $D = 0$ and this inverse formula is inapplicable, consistent with the failure of unique parameter identification.

Exercise 6.6.17: Centering the predictor makes the columns orthogonal

(a) Rewrite the actual data from Example 1.

Their mean abscissa is $(2+5+7+8)/4 = 5.5$. The transformed data are

$$(-3.5, 1), \quad (-0.5, 2), \quad (1.5, 3), \quad (2.5, 3).$$

The design matrix has columns $\mathbf{1}$ and $x^* = (-3.5, -0.5, 1.5, 2.5)^T$. Their dot product is the sum of x^*, which is zero, proving orthogonality.

(b) Solve the diagonal normal equations.

Here $X^TX = \operatorname{diag}(4, 21)$ and $X^Ty = (9, 7.5)^T$. Hence $\widehat{\beta}_0 = 9/4$ and $\widehat{\beta}_1 = 5/14$.

Conclusion. The centered fitted line is $y = 9/4 + (5/14)x^*$, with $x^* = x - 5.5$. Its uncentered form is $y = 2/7 + (5/14)x$.

Exercise 6.6.18: A general centered design has a diagonal Gram matrix

The two off-diagonal Gram entries are both $\sum_i x_i$, which is zero by the centered-coordinate assumption. Therefore

$$X^TX = \begin{bmatrix} n & 0 \\ 0 & \sum_i x_i^2 \end{bmatrix}.$$

If every centered abscissa is zero, the second diagonal entry vanishes; diagonal does not by itself mean invertible.

Exercise 6.6.19: Use an orthogonal decomposition of the observed vector

Write $y = X\widehat{\beta} + r$. The normal equations give $X^Tr = 0$, so r is orthogonal to $\operatorname{Col} X$, which contains $X\widehat{\beta}$. The Pythagorean theorem therefore applies:

$$\|y\|^2 = \|X\widehat{\beta}\|^2 + \|r\|^2.$$

With the problem's notation, this is $\mathrm{SS}(T) = \mathrm{SS}(R) + \mathrm{SS}(E)$. The zero-mean convention lets the total be interpreted as centered variation; the displayed norm identity itself follows from orthogonality.

Exercise 6.6.20: A computational formula for the fitted and residual sums

Using the normal equations in the middle step,

$$\|X\widehat{\beta}\|^2 = \widehat{\beta}^TX^TX\widehat{\beta} = \widehat{\beta}^TX^Ty.$$

Subtract this from the Pythagorean identity in Exercise 19 to obtain

$$\boxed{\mathrm{SS}(E) = y^Ty - \widehat{\beta}^TX^Ty}.$$

This avoids explicitly summing every residual square once the fit has been computed. In finite-precision arithmetic direct residual norms can be more stable when this subtraction nearly cancels two large numbers.